

Intended Learning Objectives

In this lab you will:

- Design and simulate a common-drain amplifier.
- Learn how to generate and use design charts.
- Learn how to do AC, DC, and transient simulations of a CD amplifier.
- Investigate the ringing and peaking problem in a capacitive-loaded CD amplifier with a large signal source resistance (R_{sig}) and learn how to solve it.
- Use PMOS input transistor to avoid body effect in a CD amplifier.

PART 1: Sizing Chart

1. We can show that the intrinsic gain of a MOSFET is given by

$$|A_v| \approx g_m r_o = \frac{2I_D}{V_{ov}} \times \frac{V_A}{I_D} = \frac{2V_A}{V_{ov}}$$

Interestingly, the gain only depends on λ and V_{ov} . However, to derive this expression we used $g_m = \frac{2I_D}{V_{ov}}$ which is based on the square-law. For a real MOSFET, if we compute V_{ov} and $\frac{2I_D}{g_m}$ they will not be equal. Let's define a new parameter called V-star (V^*) which is calculated from actual simulation data using the formula

$$V^* = \frac{2I_D}{g_m} \leftrightarrow g_m = \frac{2I_D}{V^*}$$

For a square-law device, $V^* = V_{ov}$, however, for a real MOSFET they are not equal. The actual gain is now given by

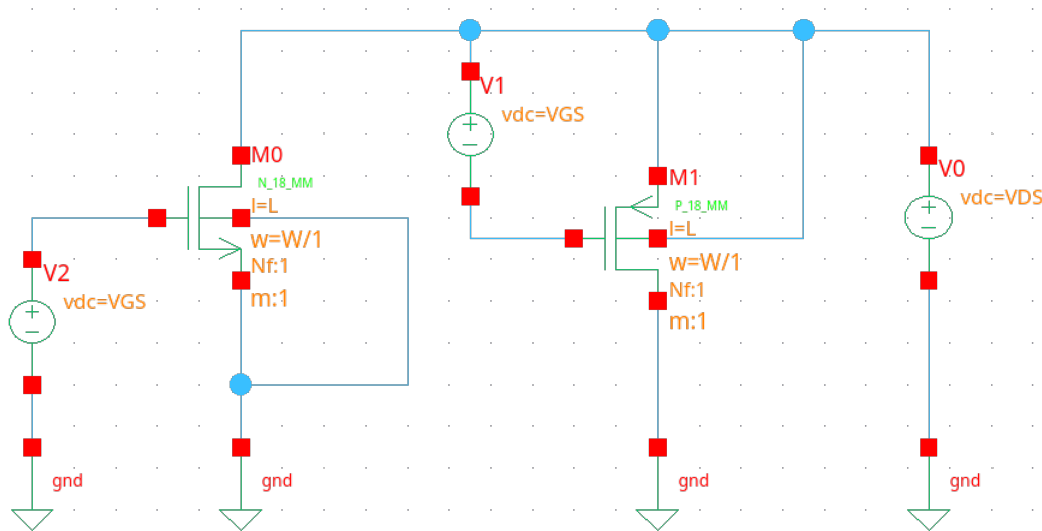
$$|A_v| \approx \frac{2V_A}{V^*}$$

The lower the V^* the higher the gain, but the larger the area and the lower the speed. An often used sweet-spot that provides good compromise between different trade-offs is $V^* = 200\text{mV}$.

2. We want to design CD and cascode amplifiers with the parameters below.

Parameter	Value
Input MOSFET	PMOS
L	1 μm
V^*	200 mV
Supply voltage	1.8 V
Current Consumption	10 μA

3. The remaining variable in the design is to calculate W . Since the square-law is not accurate, we cannot use it to determine the sizing. Instead, we will use a sizing chart generated from simulation. Create a testbench for PMOS transistor as shown below (we will use PMOS only in this lab). Use $W = 10\mu\text{m}$ (we will understand why shortly) and $L = 1\mu\text{m}$ (the same L selected before).



4. Sweep V_{GS} from 0 to $\approx V_{TH} + 0.4\text{V}$ with 10mV step. Set $V_{DS} = V_{DD}/2$.
5. We want to compare $V^* = 2I_D/g_m$ and $V_{ov} = V_{GS} - V_{TH}$ by plotting them overlaid. Use the calculator to create expressions for V^* and V_{ov} . You can save the expressions to reuse them later.
6. Plot V^* and V_{ov} overlaid vs V_{GS} . Make sure the y-axis of both curves has the same range. You will notice that at the beginning of the strong inversion region, V^* and V_{ov} are relatively close to each other (i.e., square-law is relatively valid). For deep strong inversion (large V_{ov} : velocity saturation and mobility degradation) or weak inversion (near-threshold and sub-threshold operation) the behavior is quite far from the square-law.
7. An often used sweet-spot that provides good compromise between different trade-offs is $V^* = 200\text{mV}$. On the V^* and V_{ov} chart locate the point at which $V^* = 200\text{mV}$. Find the corresponding V_{ovQ} and V_{GSQ} .
8. Plot I_D , g_m , and g_{ds} vs V_{GS} . Find their values at V_{GSQ} . Let's name these values I_{DX} , g_{mX} , and g_{dsX} .
9. Now back to the assumption that we made that $W = 10\mu\text{m}$. This is not the actual value that we will use for our design. But the good news is that I_D is always proportional to W irrespective of the operating region and the model of the MOSFET (regardless square-law is valid or no). Thus, we can use ratio and proportion (cross-multiplication) to determine the correct width at which the current will be $I_{DQ} = 10\mu\text{A}$ as given in the specs. Calculate W as shown below.

W	I_D
$10\mu\text{m}$	$I_{DX} @ V_Q^*$ (from the chart)
?	$I_{DQ} = 10\mu\text{A}$ (from the specs)

10. Now we are almost done with the design of the amplifier. Note that g_m is also proportional to W as long as V_{ov} is constant. On the other hand, $r_o = 1/g_{ds}$ is inversely proportional to W (I_D) as long as L is constant. Before leaving this part, calculate g_{mQ} and g_{dsQ} using ratio and proportion (cross-multiplication).

PART 2: CD Amplifier

1. OP (Operating Point) Analysis

- (a) Create a new schematic for the CD amplifier (the schematic is not included in the lab document and is left for the student as an exercise). Use a PMOS and use a $10\mu A$ ideal current source for biasing (note that the current source will be connected to the source terminal). Connect the source to the bulk. Use $L = 1\mu m$ and W as determined in Part 1. Use $C_L = 2pF$, $R_{sig} = 2M\Omega$, and a DC input voltage = $0V$.
- (b) Simulate the DC OP point. Report a snapshot showing the following parameters for PMOS in addition to DC node voltages clearly annotated.

Parameter	PMOS
I_D	
V_{GS}	
V_{DS}	
V_{TH}	
V_{DSAT}	
g_m	
g_{ds}	
g_{mb}	
C_{db}	
C_{gd}	
C_{gs}	
C_{sb}	
Region	

- (c) Check that the transistor operates in saturation.

2. AC Analysis

- (a) Perform AC analysis (1Hz:10GHz, logarithmic, 20points/decade) to investigate the frequency domain peaking.
- (b) Report the Bode plot magnitude.
- (c) Do you notice frequency domain peaking?
- **Cadence Hint:** Use the following expression to calculate the peaking in dB: $y_{max}(dB20(VF('vout')))$
- (d) Analytically calculate quality factor (use approximate expressions). Is the system underdamped or overdamped?
- (e) (Optional) Perform parametric sweep: $C_L = 2p, 4p, 8p$.

- Report Bode plot magnitude overlaid on same plot.
- Report the peaking vs C_L .
- Comment.

(f) (Optional) Perform parametric sweep: $R_{sig} = 20k, 200k, 2M$.

- Report Bode plot magnitude overlaid on same plot.
- Report the peaking vs R_{sig} .
- Comment.

3. Transient Analysis

(a) Use a pulse source (pulse_v_source) as your transient stimulus and set it as follows (delay = $2\mu s$, initial = 0V, period = $8\mu s$, pulse_value = 100mV, t_fall = 1ns, t_rise = 1ns, width = $4\mu s$). Run transient analysis (max step = 10n) for $10\mu s$ to investigate the time domain ringing.

(b) Report V_{in} and V_{out} overlaid vs time.

(c) Calculate the DC voltage difference (DC shift) between V_{in} and V_{out} .

- What is the relation between the DC shift and V_{GS} ?
- How to shift the signal down instead of shifting it up?

(d) Do you notice time domain ringing?

► **Cadence Hint:** Use the overshoot function to calculate the maximum overshoot as a percentage

(e) (Optional) Perform parametric sweep: $C_L = 2p, 4p, 8p$.

- Report V_{out} vs time overlaid on same plot.
- Report the overshoot vs C_L .
- Comment.

(f) (Optional) Perform parametric sweep: $R_{sig} = 20k, 200k, 2M$.

- Report V_{out} vs time overlaid on same plot.
- Report the overshoot vs R_{sig} .
- Comment.

4. Z_{out} (Inductive Rise) (optional)

(a) We want to simulate the CD amplifier output impedance. Replace C_L with an AC current source with magnitude = 1. Remove the AC input signal.

(b) Perform AC analysis (1Hz:10GHz, logarithmic, 10points/decade). The voltage across the AC current source is itself the output impedance.

(c) Plot the output impedance (magnitude and phase) vs frequency. Do you notice an inductive rise? Why?

(d) Does Z_{out} fall at high frequency? Why?

Hint: C_{gd} appears in parallel with R_{sig} .

- (e) Analytically calculate the zeros, poles, and magnitude at low/high frequency for Z_{out} . Compare with simulation results in a table.

5. How to solve the peaking/ringing problem? (optional)

- (a) Place the input/output poles away from each other (as we did when we swept C_L and R_{sig}).
- (b) (This part is optional) A compensation network can be used to compensate for the negative input impedance and prevent overshoots. Read [Johns and Martin, 2012] Section 4.4 and try to implement the compensation network.

Lab Summary

- In Part 1 you learned
 - How to generate and use design charts for NMOS and PMOS transistors.
 - How to design a PMOS common-drain amplifier.
 - How the overdrive voltage of a MOS transistor deviates from the square law in different regions of operation.
- In Part 2 you learned
 - How to do AC, DC and transient simulations of a CD amplifier.
 - How the peaking in the frequency response of a CD amplifier changes with the load capacitor and source resistance.
 - How the ringing in the transient response of a CD amplifier changes with the load capacitor and source resistance.
 - How the output impedance of a CD amplifier shows inductive behavior.